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Dear Prof Mak,

I have tried to draft the manuscript of the scoping review:

1. There is a similar review published recently in 2024: **The Association Between Chronic Tobacco Smoking and Brain Alterations in Schizophrenia: A Systematic Review of Magnetic Resonance Imaging Studies** (https://doi.org/10.1093/schbul/sbae088).
2. The article includes many neuroimaging studies. As a preliminary search, I have scanned through all the primary studies included in this systematic review and created data extraction tables for a summary (excel file for general study characteristics; word file for neuroimaging conclusion; see attached).
3. I have not yet searched the database again for any further primary studies. But I hope that the summary derived from the studies extracted can help a little… Details of the content have been included as much as possible (except for results and discussion here).
4. The original focus for extracting neuroimaging results from studies is the comparative results between smoking and non-smoking patients with schizophrenia. However, the participants among the studies vary a lot, so I think it is possible to also report differences between smoking schizophrenia patients with any of the following groups: (1) non-smoking schizophrenia patients, (2) smoking healthy controls and (3) non-smoking healthy controls.

I am not sure about the correct structure of a review. And I have tried my best to make it a proper writing! Thank you very much!

Best regards,
Toby
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# The Relationship Between Structural Brain Changes and Psychiatric Symptoms Severity  Between Patients with Schizophrenia Who Smoke and Those Who Do Not Smoke: A Scoping Review


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## Abstract

*Background.*

*Objectives.*

*Methods.*

*Results.*

*Conclusions.*


## Introduction

Schizophrenia is a severe psychiatric disorder characterised by positive symptoms (e.g., hallucinations, delusions), negative symptoms (e.g., avolition, social withdrawal), and cognitive impairments (Andreasen et al., 1994; McCutcheon et al., 2023). Although the presence of positive symptoms often draws clinical attention to the patients, negative and cognitive symptoms that are  more strongly associated with reduced medication adherence, and poorer long-term functional outcomes (Carbon & Correll, 2014; Correll et al., 2024). With a lifetime prevalence of approximately 0.4%  (Saha et al., 2005), schizophrenia significantly diminishes quality of life across physical, psychological, and social domains (Dong et al., 2019). Despite extensive research, prognostic improvements for affected individuals remain limited (Molstrom et al., 2022).

Smoking rates among people with schizophrenia (60%) vastly exceed those in the general population, and are with a heavier nicotine dependence, lower quit rates, and earlier smoking initiation (Šagud et al., 2018; Gogos et., 2019). The underlying mechanism is multifaceted, including genetic, neurobiological, and psychosocial factors. An overlap in genetic variants and a convergence on neurotransmitter systems between nicotine addiction and psychosis have been observed (Leonard et al., 2007; Lucatch et al., 2018). The proposed self-medication hypothesis suggested that nicotine transiently normalises brain dysfunction, with short-term benefits on attention and possibly positive symptoms (Isuru & Rajasuriya, 2019). However, evidence to support the long-term cognitive enhancement by smoking throughout the course of disease is insufficient. In schizophrenia, smoking increases the already elevated cardiovascular risk, leading to higher standardized mortality, more hospitalizations and greater symptom severity (Kobayashi et al., 2010; Kelly et al., 2011; Uludag & Zhao, 2023). In addition, accelerated metabolism of antipsychotics by nicotine can result in a need for higher doses to achieve therapeutic effect (Lyon, 1999). Currently, the literature on how smoking affects schizophrenia is mixed, with inconsistencies often due to small samples, selection biases, variability in defining smoking exposure, and inadequate control for confounders such as medication, illness stage, and comorbid substance use.

Neuroimaging techniques, including structural and functional magnetic resonance imaging MRI), provide an objective means to investigate the neural substrates of both schizophrenia and nicotine use. Initially used to rule out organic causes for psychotic symptoms (Keshavan et al., 2020), neuroimaging now robustly characterizes patterns of macrostructural, microstructural, functional, and neurochemical alteration in the disorder. Advanced analytic methods, including deep learning and artificial intelligence, are further enhancing thesensitivity of these techniques to detect subtle neural patterns (Du et al., 2025). While prior reviews have explored the neural correlates of schizophrenia and chronic smoking separately (Koster et al., 2025), a focused synthesis on their intersection is needed.

There is a clear need to systematically map what is currently known about neuroimaging differences between people with schizophrenia who smoke and those who do not. A scoping review is necessary due to various imaging modalities, diverse analytic approaches, and heterogeneous clinical samples. It can highlight consistencies and discrepancies across modalities, brain regions, and analytic strategies, and identify methodological gaps. It can also connect seemingly contradicting interpretations, for example, effects due to acute nicotinic modulation, selection effects, medication interactions, neurotoxic impacts of chronic smoking or neuroprotective signals.

Therefore, this scoping review aims to: (1) describe the neuroimaging differences between patients with schizophrenia who smoke and those who do not, across different imaging modalities; and (2) synthesize the study designs and analytic approaches used to investigate these differences. By integrating these findings, we hope to generate novel insights into how nicotine exposure interacts with the neuropathology of schizophrenia, outline possible biomarkers for targeted interventions, and help reconcile why some studies suggest neuroprotective or normalizing effects while others indicate neural harm.

This work has broader implications for clinical practice and research. Clarifying the brain-based relationship between smoking and schizophrenia can inform the design of tailored smoking cessation interventions, with the ultimate goals of improving both cardiovascular health and psychiatric outcomes. Furthermore, linking clinical phenomena to their neurobiological substrates may help reduce stigma by framing schizophrenia within a coherent neuroscience model and guide future translational work that bridges genetic risk, environmental exposures, and circuit-level dysfunction.

## Methods

*Review Design*

This scoping review was conducted in accordance with the Joanna Briggs Institute (JBI) methodology for scoping review (Peters et al., 2020). Our protocol was drafted using the Preferred Reporting Items for Systematic Reviews and Meta-analysis Protocols (PRISMA-PXXXXXX), which was revised by XXX, and was disseminated through XXX to solicit additional feedback. The final protocol was registered prospectively with XXX on XXX.

*Eligibility Criteria (PCC Framework)*

Our eligibility criteria were defined using the PCC (Participants, Concept, Context) framework recommended by JBI (Tricco et al., 2020). The review considered studies including adult **participants** (aged 18 and above) from both clinical (i.e. individuals with a diagnosis of schizophrenia, schizoaffective disorder, or schizophreniform disorder based on standardized criteria (DSM-IV or ICD), and non-clinical (e.g. general population, healthy controls) populations. Studies focusing exclusively on adolescents or paediatric populations were excluded.

The core **concept** of this review is the interplay between smoking behaviors, structural brain integrity as measured by sMRI, and psychiatric symptom severity. This includes smokers were preferentially defined as current daily smokers for at least one year (current smoker), former, and never smokers. Due to varied definitions of smoking across studies, we accepted other study-specific criteria with explicit classification. Studies were required to report neuroimaging-derived brain differences that compared patients who smoke with non-smoking patients and/or healthy smokers or healthy non-smokers. Specifically, metrics derived from T1-weighted structural MRI (sMRI), including but not limited to reginal brain volumes (e.g. cortical and subcortical gray matter), cortical thickness, and surface area. All neuroimaging modalities were considered for the synthesis of data, such as structural measures (for example, gray matter volume, cortical thickness, white matter integrity, diffusion metrics) and functional measures (for example, task-based activation, resting-state connectivity, and network indices). Additionally, psychiatric symptom severity was measured with quantified standardised clinical scales such as PANSS or BPRS for psychosis.

Context: We included published primary research, peer-reviewed and grey literature, and both cross-sectional and longitudinal studies that reported baseline correlational data. . When systematic reviews were identified, we screened their included studies to ensure that eligible primary studies were not missed. Only studies in English were included.

*Information sources and search strategy*

A preliminary search was conducted in PubMed, Embase, and PsycINFO via ProQuest to search papers from inception to February 2025, using the strategy: (schizophrenia OR psychosis OR psychoses OR “psychotic disorder” OR “schizophrenic disorder”) AND (smoking OR tobacco OR cigarette OR nicotine) AND (neuroimaging OR “brain imaging”). 17 records have been identified. Full search strategies were drafted by XXX and are provided in an additional file. Grey literature was searched in XXX, and reference lists of relevant reviews were scanned to identify additional studies. Final search results were imported into EndNote, and XXX removed duplicates. The comprehensive search includes six databases: MEDLINE and CINAHL Complete via EBSCOhost; Embase and PsycINFO via ProQuest; Web of Science; and the Cochrane Library. (NOT YET SEARCHED)

*Selection of sources of evidence*

To ensure consistency, XXX publications were initially co-screened by the review team, with findings discussed and the data extraction manual refined before full screening. XXX reviewers (XXX) independently screened titles and abstracts, followed by full-text screening for potentially eligible records. Disagreements were discussed and resolved by consensus.

*Data charting process*

A standardized data charting form was jointly developed by XXX reviewers to define variables and was iteratively refined after pilot searching. Reviewers independently extracted data from eligible studies using the XXX web software. Any disagreements were resolved through discussion among XXX reviewers. Extracted domains included participant characteristics, clinical information, smoking status and nicotine dependence, neuroimaging methods, and main findings.

*Data items*

For each study, we collected author and year, country or setting, and general characteristics of participants (gender, age, education level, IQ, and ethnicity where reported). Clinical characteristics included diagnosis and subtype, illness duration, medication type and dose, and symptom severity scales. Possible confounding factors such as substance use history, including cannabis use were also included. Smoking characteristics included the definition of smoking, duration of smoking, , pack-years, cigarettes consumed per day and FTND scores. Neuroimaging details included modality, type of scanner, analytical approaches (region-of-interest versus whole-brain), and statistical methods. Neuroimaging findings were reported in structural, diffusion, functional, spectroscopic, and molecular domains.

*Synthesis of results*

Across all imaging modalities, we summarized participant profiles and smoking measures. A narrative synthesis was conducted, grouping studies by neuroimaging modality and reporting consistencies and divergences of principal neuroimaging conclusions across studies. We also mapped gaps in the evidence base, including understudied regions and circuits and methodological limitations such as small sample sizes and confounding by medication, illness stage, and comorbidities, to inform future regions of interest and study design recommendations.

## Results

Selection of sources as evidence

*(PRISMA Flow diagram inserted)*

Characteristics of sources of evidence

*(Description needed)*

Results of Individual sources of evidence

*Table 1 (participant information, clinical information, smoking details) inserted*

Synthesis of results

*Table 2 (neuroimaging details & main findings) inserted*

## Discussion and Funding

*Summary of evidence*

*Limitations*

*Conclusions*

*Funding*


- MHRC, HKPolyU – Project Title: Using Functional Magnetic Resonance Imaging (fMRI) to assess the effects of Acceptance and Commitment Therapy for smoking cessation and reduction of psychotic symptoms in patients with schizophrenia (Grant Code: 1-BBCL)
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